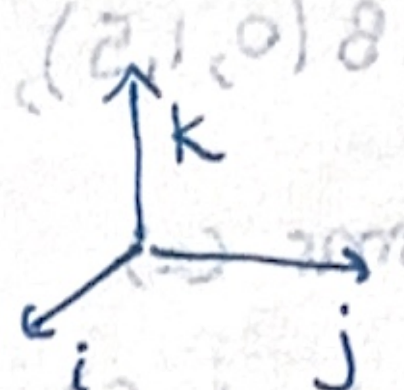
Seminar 4 - Warm-up

$\mathcal{K} = (O, \mathcal{B}), \ \mathcal{B} = (i, j, k).$



4.1. $a = i + 2j - 2k, \ b = 7i + 4j + 6k; \quad a \times b = ?$

$$a \times b = (i + 2j - 2k) \times (7i + 4j + 6k)$$

$$= 7\underbrace{i \times i}_{0} + 4\, i \times j + 6\, i \times k + 14\, j \times i + 8\underbrace{j \times j}_{0} + 12\, j \times k - 14\, k \times i - 8\, k \times j - 12\underbrace{k \times k}_{0}$$

$$= -10\underbrace{i \times j}_{k} + 20\underbrace{i \times k}_{-j} + 20\underbrace{j \times k}_{i}$$

$$= 20i - 20j - 10k$$

$$a \times b = \begin{vmatrix} i & j & k \\ 1 & 2 & -2 \\ 7 & 4 & 6 \end{vmatrix} = 20i - 20j - 10k.$$

4.2. $A(1,1,0), \ B(1,2,0), \ C(0,y,0), \ y \in \mathbb{R}.$

? $A_{\Delta ABC} \to$ independent of y.

? $\mathbb{E}^2$

$$A_{ABC} = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

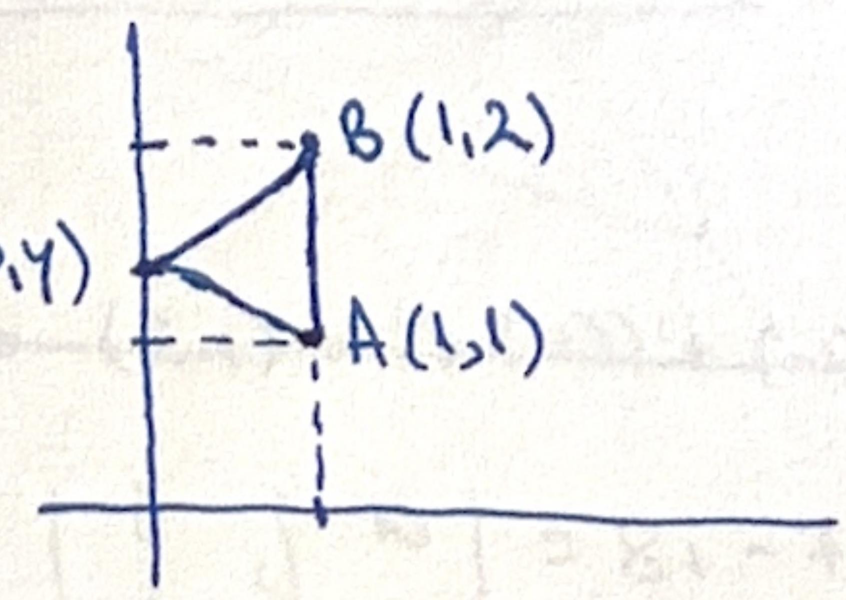
$$\vec{AB} = B - A = (0,1,0) \Rightarrow \vec{AB} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\vec{AC} = C - A = (-1, y-1, 0) \Rightarrow \vec{AC} = \begin{bmatrix} -1 \\ y-1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 0 & 1 & 0 \\ -1 & y-1 & 0 \end{vmatrix} = k \cdot \begin{vmatrix} 0 & 1 \\ -1 & y-1 \end{vmatrix} = k \Rightarrow \text{indep. of } y.$$

The length of $\vec{AB} \times \vec{AC}$ is indep. of $y \Rightarrow$ Area ΔABC is also indep. of y.

$\mathbb{E}^2$:



Area $(ABC) = \frac{1}{2} |AB| \cdot h_C$

$h_C = d(C, AB)$ which does not depend on y because C moves parallel to AB.

4.3. $a(2,-3,1), \ b(-3,1,2), \ c(1,2,3)$

$(a \times b) \times c = ?$

$a \times (b \times c) = ?$

$$\langle a, c \rangle = (a_x \cdot c_x) + (a_y \cdot c_y) + (a_z \cdot c_z) = (2 \cdot 1) + (-3 \cdot 2) + (1 \cdot 3) = 2 - 6 + 3$$

$$(\underline{a} \times \underline{b}) \times \underline{c} = \langle \underline{a} \cdot \underline{c} \rangle \underline{b} - \langle \underline{b} \cdot \underline{c} \rangle \underline{a} = (2 - 6 + 3)b - (-3 + 2 + 6)a = -b - 5a$$

$$= -\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} - 5 \cdot \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} - \begin{bmatrix} 10 \\ -15 \\ 5 \end{bmatrix} = \begin{bmatrix} -7 \\ 14 \\ -7 \end{bmatrix}$$

$$a \times (b \times c) = \langle a \cdot c \rangle b - \langle a \cdot b \rangle c = -\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} + 7 \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 13 \\ 19 \end{bmatrix}$$

(4.4) $A(1,2,-1)$, $B(0,1,5)$, $C(-1,2,1)$, $D(2,1,3)$ coplanar?

A,B,C,D coplanar $\Leftrightarrow$ $\vec{AB}, \vec{AC}, \vec{AD}$ $\boxed{\text{lin. dependent.}}$

$\vec{AB} = B - A = (-1,-1,6)$

$\vec{AC} = C - A = (-2,0,2)$

$\vec{AD} = D - A = (1,-1,4)$

$$\vec{AB} = \begin{bmatrix} -1 \\ -1 \\ 6 \end{bmatrix}, \quad \vec{AC} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}, \quad \vec{AD} = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}.$$

Volume of tetrahedron: $ABCD = 0$. $\Leftrightarrow$ A,B,C,D coplanar.

$$\Leftrightarrow \begin{vmatrix} -1 & -1 & 6 \\ -2 & 0 & 2 \\ 1 & -1 & 4 \end{vmatrix} = 0 \quad \text{true.}$$

(4.5.) $A(2,-1,1)$, $B(5,5,4)$, $C(3,2,-1)$, $D(4,1,3)$.

a) $V_{ABCD} = ?$

b) common $\perp$ of sides AB and CD.

a) $$Vol = \left|\frac{1}{6}[\vec{AB}, \vec{AC}, \vec{AD}]\right| = \left|\frac{1}{6} \cdot \begin{vmatrix} 3 & 6 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & 2 \end{vmatrix}\right| = |-3| = 3.$$

$$\vec{AB} = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}, \quad \vec{AC} = \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}, \quad \vec{AD} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \quad \vec{CD} = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

b) $$\left.\begin{array}{l} AB: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}}_{= A} + t \cdot \underbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}_{= \vec{AB}/3} \\[2em] CD: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}}_{= C} + s \cdot \underbrace{\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}}_{= \vec{CD}} \end{array}\right\} \text{parametric eq. of the lines.}$$

$$\frac{\vec{AB}}{3} \times \vec{CD} = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 1 & -1 & 4 \end{vmatrix} = i \cdot \begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} - j \cdot \begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} + k \cdot \begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} =$$

$$= i \cdot (8+1) - j \cdot (4-1) + k(-1-2) = 9i - 3j - 3k = 3(3i - j - k)$$

$$= 3 \cdot \underbrace{\begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}}_{= n}$$

Common $\perp$: $$\begin{cases} \begin{vmatrix} x - x_A & y - y_A & z - z_A \\ 1 & 2 & 1 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Leftrightarrow -1(x-2) + 4(y+1) - 7(z+1) = 0 \\[2em] \begin{vmatrix} x - x_C & y - y_C & z - z_C \\ 1 & -1 & 4 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Leftrightarrow 5(x-3) + 13(y-2) + 2(z+1) = 0 \end{cases}$$

$$(=) \begin{cases} -x+4y-7z+13=0 \\ 5x+13y+2z-39=0. \end{cases}$$

4.6. $a(2,3,-1)$, $b(1,-1,3)$.

$p \Rightarrow$ orthogonal to $\underline{a}$ and $\underline{b}$

$\langle p, 2i-3j+4k \rangle = 51$

(M1): $v(x,y,z)$ arbitrary vector.

impose: $v \in \langle a,b \rangle^{\perp} \iff v \perp a$ and $v \perp b$

$\iff v \cdot a = 0$ and $v \cdot b = 0$

$$\iff \begin{cases} \cancel{2x-3y+4}\ 2x+3y-z=0 \Rightarrow z = 2x+3y. \\ x-y+3z=0 \Rightarrow y = x+3z. \end{cases}$$

$z = 2x+3(x+3z) = 5x+6z \Rightarrow -5x = 5z \Rightarrow x=-z.$

$\Rightarrow \underline{z=-x}$

$\Rightarrow y = x+3\cdot z = x+3\cdot(-x) = -2x$

$\Rightarrow \underline{y=-2x}.$

$\Rightarrow v \in \langle a,b\rangle^{\perp} \iff v = v(x, y(x), z(x)).$

$= v(x,-2x,-x).$

(M2): $\boxed{\langle a,b\rangle^{\perp} = \langle a \times b \rangle} = \left\langle \begin{bmatrix} 8 \\ -7 \\ -5 \end{bmatrix} \right\rangle = \{(8t,-7t,-5t) : t \in \mathbb{R}\}.$

$\cancel{\langle a \times b \rangle = (2\cdot 1)+(3\cdot(-1))+(-1)\cdot 3 =}$

$$a \times b = \begin{vmatrix} i & j & k \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} = 8i-7j-5k.$$

$p \in \langle a,b\rangle^{\perp} \iff p = t\cdot\begin{bmatrix} 8 \\ -7 \\ -5 \end{bmatrix}$ for some $t \in \mathbb{R}$.

? $p\cdot(2i-3j+4k)=51 \iff (t\cdot 8i - t\cdot 7j - t\cdot 5k)\cdot(2i-3j+4k)=51$

$\iff 16t+21t-20t=51$

$\iff t = \frac{51}{17} = 3.$

4.7. $A(1,1,0)$, $B(1,0,1)$, $C(+1,-1,-1)$, $D(0,y,z)$.
$y,z \in \mathbb{R}$.

$V_{ABCD} \to$ indep. on y,z.

· $Vol_{ABCD} = \frac{1}{6} \left| [\vec{AB}, \vec{AC}, \vec{AD}] \right|$

· $\vec{AB} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$, $\vec{AC} = \begin{bmatrix} 0 \\ -2 \\ -1 \end{bmatrix}$, $\vec{AD} = \begin{bmatrix} -1 \\ y-1 \\ z \end{bmatrix}$

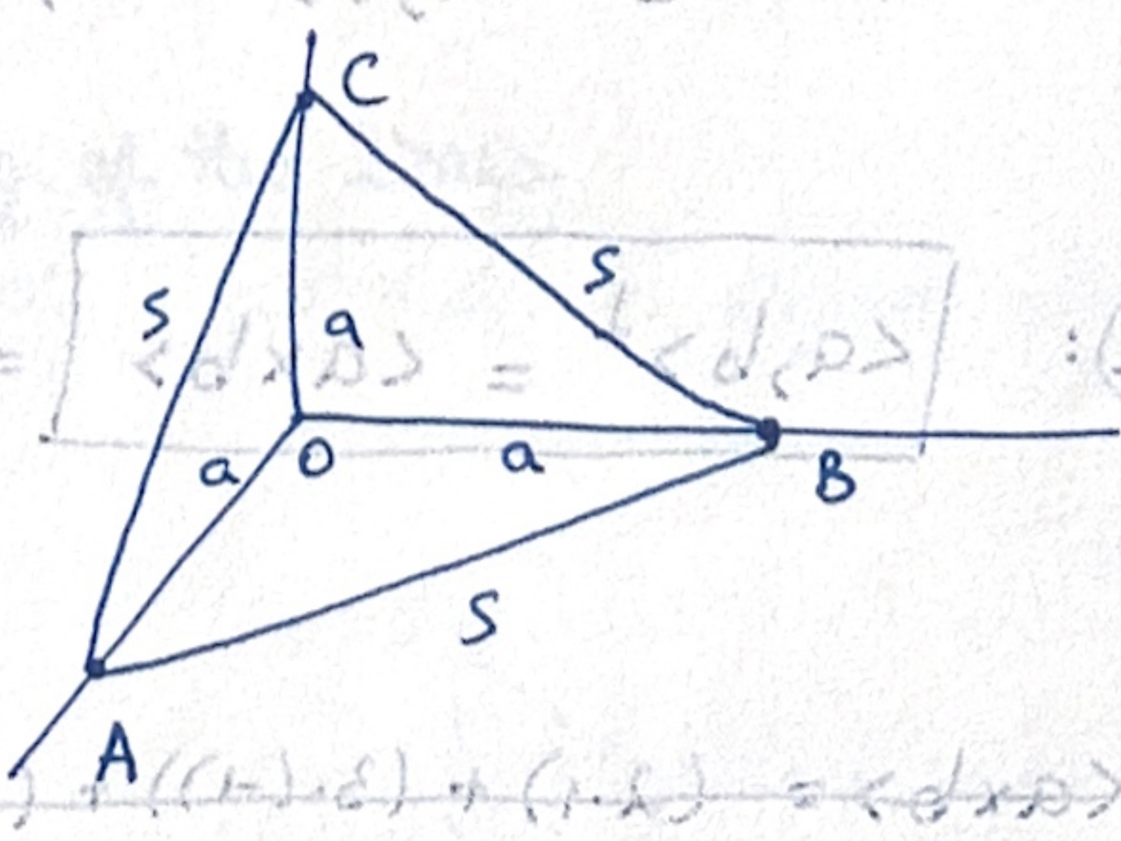
· $[\vec{AB}, \vec{AC}, \vec{AD}] = \begin{vmatrix} 0 & -1 & 1 \\ 0 & -2 & -1 \\ -1 & \cancel{y-1} & \cancel{z} \end{vmatrix} = -1 \cdot \begin{vmatrix} -1 & 1 \\ -2 & -1 \end{vmatrix} = -3.$

$\Rightarrow Vol_{ABCD} = \frac{1}{6} \cdot |-3| = \frac{3}{6} = \frac{1}{2}$. constant / indep. on y and z

Obs! $Vol_{ABCD} = \underbrace{\frac{1}{3} \cdot Area(ABC)}_{\text{indep. of } y,z} \cdot h_D \to$ volume of pyramid with base ABC.
↳ $d(D, ABC)$, but D moves in a plane parallel to ABC. $\Rightarrow$ dist is const.

? 4.8. π plane
$(1,1,1)$ normal vector intersecting the coord. axes in A,B,C.

$V_{OABC} \to h = d(O, \pi)$
$a = |OA|$
$s = |AB|$



Assume $A(a,0)$, i.e. $a > 0$

? $\pi: \frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$

$(=)\ x + y + z - a = 0$

$h = d(O, \pi) = \frac{|a|}{\sqrt{3}} = \frac{a}{\sqrt{3}} \Rightarrow a = \sqrt{3} \cdot h$.

$S = \sqrt{2} a \Rightarrow a = \frac{S}{\sqrt{2}}$.

$V = Vol_{OABC} = \frac{1}{6}$ volume cube with side a.

$\Rightarrow V = \frac{a^3}{6} = \frac{\sqrt{3}}{2} \cdot h^3 = \frac{s^3}{12\sqrt{2}}$ $\quad s/\sqrt{6}$

Heron check: $A_{ABC} = \frac{\sqrt{3}}{4} s^2 \Rightarrow V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4} \cdot s^2 \cdot h = \frac{s^3}{12\sqrt{2}}$

4.9. $\phi : \mathbb{V}^3 \to \mathbb{V}^3$, $\phi(v) = (2i-j) \times v$. $\to$ linear map. its matrices?

$$\phi(i) = (2i-j)\times i = \underbrace{2i\times i}_{0} - j\times i = i \times j = k.$$

$$\phi(j) = (2i-j)\times j = 2i \times j = 2k$$

$$\phi(k) = (2i-j)\times k = 2i\times k - j \times k = -2j - i.$$

$$\Rightarrow [\phi] = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \phi(i) & \phi(j) & \phi(k) \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & -2 \\ 1 & 2 & 0 \end{bmatrix}$$

4.10. $u = -i + 3j + k$

$w = k$

$\phi : \mathbb{V}^3 \to \mathbb{R}$, $\phi(v) = [v, u, w]$. matrix?

$$\phi(i) = \begin{vmatrix} 1 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} = 3$$

$\downarrow \quad \downarrow \quad \downarrow$

$i \quad (j) \quad (k)$

$$\phi(j) = \begin{vmatrix} 0 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ -1 & 3 \end{vmatrix} = -(-1) = 1.$$

$$\phi(k) = \begin{vmatrix} 0 & 0 & 1 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$\Rightarrow [\phi] = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \phi(i) & \phi(j) & \phi(k) \\ \downarrow & \downarrow & \downarrow \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \end{bmatrix}.$$